

VOLUME II - EXISTING SOLUTIONS / DOCUMENT 2 OF 3

Technology Comparison

Structured Evaluation of Candidate Sensing, Positioning, Communication and Compute Technologies

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ABSTRACT

This document evaluates the candidate technologies for each functional layer of the proposed system against weighted criteria derived from the field requirements. It covers perception modalities, positioning methods, communication bearers, compute platforms, human-machine interface options and visibility sensing. The evaluation applies a common scoring framework, states the physics-based reasoning behind each score, and concludes with a recommended technology set and the rationale for each selection, including the technologies deliberately rejected.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Evaluation Framework

Each candidate is scored 1 to 5 against weighted criteria. Weights derive from the constraint register of Volume I Document 2 and the requirements of Volume I Document 3.

Criterion	Weight	Rationale for weight
Fog performance at MOR 3-20 m	25%	The defining constraint. A technology that fails here cannot be a primary layer regardless of other merits.
Required range and resolution	20%	The stopping distance analysis makes range a hard gate.
Environmental ruggedness	15%	Ore mud, vibration, humidity and heat have historically killed mining pilots.
Maturity and field proof	15%	This is a production mine, not a research platform.
Cost per vehicle at fleet scale	15%	A large mixed fleet makes unit cost a first-order constraint.
Integration and retrofit effort	10%	Brownfield mixed-fleet retrofit with short installation windows.

Scores are: **5** excellent, **4** good, **3** adequate, **2** marginal, **1** unsuitable.

2. Layer 1 - Perception Modalities

2.1 Physics-based screening

Before scoring, the modalities are screened on physics, because at this operating point physics eliminates candidates outright rather than merely disadvantaging them.

Attenuation in fog is governed by Mie scattering, whose efficiency peaks where particle size is comparable to wavelength. Fog droplets are typically 1-20 μm in diameter. The size parameter $x = \pi \cdot d / \lambda$ determines the regime: for $x \gg 1$ scattering is strong (geometric/Mie regime); for $x \ll 1$ scattering falls off steeply (Rayleigh regime, where scattering cross-section scales as the sixth power of diameter over the fourth power of wavelength).

Modality	Wavelength	Size parameter x for a 10 μm droplet	Regime	Consequence
Visible camera	0.55 μm	~ 57	Strong Mie	Blind at MOR 3-5 m
NIR LiDAR	905 nm	~ 35	Strong Mie	Range collapse; severe backscatter clutter
SWIR LiDAR	1550 nm	~ 20	Strong Mie	Degraded; modest gain over 905 nm
LWIR thermal	10 μm	~ 3	Mie, reducing	Partial penetration; useful for thermal-contrast targets
mmWave radar	3.9 mm (77 GHz)	~ 0.008	Deep Rayleigh	Essentially transparent
mmWave radar	3.8 mm (79 GHz)	~ 0.008	Deep Rayleigh	Essentially transparent
Low-THz radar	1.0 mm (300 GHz)	~ 0.03	Rayleigh	Near-transparent; higher resolution, smaller aperture
UHF RF ranging	0.3-1 m	$\sim 1\text{e-}5$	Deep Rayleigh	Unaffected
Ultrasonic	~ 8 mm (in air)	n/a - acoustic	n/a	Unaffected by fog but degraded by wind and rain; very short range

Table 2.1 - Physics screening. The three-order-of-magnitude gap between optical and millimetre-wave size parameters is the reason radar is not merely better in fog but categorically different.

Supporting measured data: at 77 GHz, attenuation in heavy rain at 25 mm/hr is reported at approximately **10 dB/km**, while a 905 nm LiDAR under comparable low-visibility conditions suffers approximately **35 dB/km**. In fog the disparity widens further, because fog droplets are far better matched to optical wavelengths than rain drops are. Simulation of a 77 GHz radar and 1550 nm laser configuration indicates detection ranges on the order of **260 m in heavy fog and 460 m in mild fog** at a 20 dB SNR threshold.

Screening conclusion. Any architecture whose *primary* forward-sensing layer is optical - visible camera, NIR LiDAR, or SWIR LiDAR - is disqualified for this application. This is not a matter of engineering effort or algorithm quality. The information does not reach the sensor. The primary layer must be millimetre-wave radar, supported by cooperative RF. Optical modalities may serve as secondary layers active in classes V0 to V2, where they contribute classification quality that radar alone cannot.

2.2 Detailed modality comparison

Millimetre-wave radar, 77 / 79 GHz

Mechanism. FMCW radar transmits a frequency-swept signal and measures range from beat frequency, velocity from Doppler shift, and azimuth (and in 4D variants, elevation) from a MIMO antenna array.

Strengths. Essentially unaffected by fog, rain, dust and smoke. Direct velocity measurement by Doppler at approximately 0.1 m/s resolution, which is decisive for closing-rate computation and for discriminating moving from static clutter. Detection ranges commonly to 200-250 m for vehicle-sized targets, exceeding LiDAR's typical 150 m. Tolerates a fouled radome far better than any optical sensor tolerates a fouled window - a mud film attenuates but does not blind. Automotive mass production has driven cost, size and ruggedness to commodity levels. The 79 GHz band offers wider bandwidth and better range resolution than 24 GHz, and smaller apertures for a given beamwidth.

Weaknesses. Lower angular resolution than LiDAR, so object shape and class are harder to infer from radar alone. Multipath and clutter in a confined pit environment with large conductive surfaces - a real concern at Bailadila given the ore body. Ground clutter on a graded, bermed road requires careful elevation handling; 4D imaging radar with elevation resolution substantially mitigates this. Poor discrimination between a small metallic object and a large weakly reflective one without tracking and classification.

Score: Fog 5 | Range 5 | Rugged 5 | Maturity 5 | Cost 5 | Integration 4 -> **Weighted 4.85**

4D imaging radar

Mechanism. High-channel-count MIMO radar providing range, azimuth, elevation and Doppler, generating a sparse point cloud rather than a target list.

Strengths. Retains all the fog performance of conventional mmWave radar while substantially improving angular resolution and adding elevation, which addresses the principal weakness above. Elevation resolution allows discrimination of an overhead structure from an in-path obstruction and improves ground clutter rejection on gradients - critical on a ramped haul road. Hexagon has already introduced 4D radar support into its mining CAS, indicating sector acceptance.

Weaknesses. Higher cost than basic automotive radar, though falling rapidly. Higher data rate and compute load. A younger technology with less field history in mining.

Score: Fog 5 | Range 5 | Rugged 5 | Maturity 4 | Cost 4 | Integration 4 -> **Weighted 4.60**

LWIR thermal imaging, 8-14 um

Mechanism. Uncooled microbolometer arrays imaging emitted thermal radiation.

Strengths. Requires no illumination, so it is unaffected by darkness, glare or headlight backscatter. Excellent contrast on thermally distinct targets - people, running engines, hot brakes and tyres - which are exactly the highest-consequence targets. Penetrates dust and smoke well. Partial fog penetration, materially better than visible.

Weaknesses. Fog penetration is **partial, not complete**. At 10 um wavelength against 10 um droplets the size parameter is approximately 3, still within the Mie regime; attenuation at MOR 3-5 m remains significant. Poor contrast on thermally neutral targets - a cold rock, a parked trailer, a wet berm after rain. Performance degrades further in high humidity because water vapour absorbs in parts of the LWIR band. A germanium window is expensive and, critically, **fouls with ore mud exactly like any other optical aperture**. No direct range measurement.

Score: Fog 3 | Range 3 | Rugged 3 | Maturity 4 | Cost 3 | Integration 3 -> **Weighted 3.10**

Verdict. A valuable **secondary** layer, particularly for personnel detection, and a strong complement in V1-V2 conditions. Not a primary layer, and the widespread assumption that "thermal sees through fog" substantially overstates its capability in dense water fog.

LiDAR, 905 nm and 1550 nm

Mechanism. Time-of-flight or FMCW ranging using pulsed or modulated laser light.

Strengths. Outstanding angular and range resolution; dense 3D point clouds; the best available geometric description of the environment. Excellent for road-edge detection, berm profiling and precise obstacle geometry in clear conditions.

Weaknesses. Severe degradation in fog. Published work indicates **LiDAR alone fails beyond approximately 70 m in dense fog**, and at MOR 3-5 m useful range collapses far below that. Backscatter from near-field droplets creates a dense false return cloud that must be filtered, consuming range gate and processing. Optical aperture fouls with ore mud and requires active cleaning. Higher cost, typically three to five times an equivalent radar.

Score: Fog 1 | Range 4 | Rugged 2 | Maturity 4 | Cost 2 | Integration 3 -> **Weighted 2.40**

Verdict. Valuable for **offline mapping and digital road model generation**, where it can be flown or driven in clear conditions to build the high-fidelity road model on which synthetic guidance depends. Not suitable as a real-time fog-condition sensing layer. This distinction - LiDAR as a mapping tool, not a perception tool - is an important and often missed architectural point.

Visible-band camera with AI

Score: Fog 1 | Range 2 | Rugged 2 | Maturity 5 | Cost 5 | Integration 4 -> **Weighted 2.65**

Verdict. Essential for V0-V2 operation, for classification quality, for event recording and investigation, and for operator familiarity. Worthless at V3-V4. Must never be a primary layer, and must never be presented to the operator in a way that implies coverage it does not have.

Ultrasonic

Score: Fog 4 | Range 1 | Rugged 3 | Maturity 5 | Cost 5 | Integration 4 -> **Weighted 3.15**

Verdict. Fog-immune but range-limited to a few metres and degraded by wind and heavy rain. Acceptable for very-short-range parking and spotting assistance. Irrelevant to the haul road problem.

Cooperative RF ranging (UWB, sub-GHz, RF time-of-flight)

Mechanism. Radio transponders on vehicles and personnel exchanging ranging and identity messages.

Strengths. Completely unaffected by fog. Does not require line of sight - the SAFEmine lineage reports 360 degree protection at any visibility with typical range of 800 m without line-of-sight dependence. Provides identity and intent, not merely presence. Low cost, low power, mature. Handles the non-line-of-sight geometry of switchbacks and ridges, where an optical or radar sensor sees only the rock face.

Weaknesses. Detects only what is equipped. Blind to rockfall, washout, stray animals, unequipped contractor vehicles and any equipment whose unit has failed. Ranging accuracy varies by technology; UWB is excellent at short range, sub-GHz RF less precise but longer range. Multipath in a confined pit degrades ranging accuracy.

Score: Fog 5 | Range 5 | Rugged 5 | Maturity 5 | Cost 5 | Integration 4 -> **Weighted 4.90**

Verdict. The **highest-scoring single technology** in this evaluation, and the natural primary layer for vehicle-to-vehicle awareness. Its single weakness - blindness to non-cooperative hazards - is precisely what the radar layer exists to cover. The two are complementary in the strongest sense: each one's blind spot is the other's strength.

Magnetic field proximity detection

Score: Fog 5 | Range 1 | Rugged 4 | Maturity 4 | Cost 4 | Integration 4 -> **Weighted 3.55**

Verdict. Fog-immune with a well-defined short-range zone; the strongest option for the pedestrian blind-zone layer specifically. Requires on-site validation given the high-grade ferrous environment. Not a haul road technology.

2.3 Perception recommendation

Layer	Technology	Role	Active in classes
Primary cooperative	RF cooperative ranging and broadcast	Vehicle and personnel identity, position, velocity, intent; non-line-of-sight	All
Primary non-cooperative	4D imaging radar, 77/79 GHz, forward and quarters	Long-range hazard detection and ranging regardless of visibility	All
Short-range blind zone	Short-range radar, supplemented by magnetic-field PDS for personnel	360 degree blind-zone protection	All
Secondary classification	LWIR thermal, forward	Personnel and hot-target detection; partial fog penetration	V0-V3
Secondary classification	Visible camera, forward and rear	Classification, recording, DGMS rear-view compliance	V0-V2
Mapping (offline)	LiDAR, survey-grade	Digital road model generation and periodic revalidation	Clear conditions only

The architecture deliberately places **two independent fog-valid layers** - cooperative RF and radar - in the primary role, with the optical layers relegated to secondary contribution. This satisfies requirement FR-11 and, importantly, means the system's core function degrades gracefully rather than catastrophically if any one layer fails or fouls.

3. Layer 2 - Positioning

Technology	Accuracy	Fog	Availability in pit	Cost	Score	Verdict
Standard GNSS	2.5-10 m	Unaffected	Degraded by shadowing	Very low	2.2	Insufficient - coarser than lane width
SBAS / GAGAN	1-3 m	Unaffected	Degraded	Low	2.8	Insufficient for lane guidance
DGPS (local base)	0.3-1 m	Unaffected	Degraded	Low-moderate	3.8	Marginal; meets PR-08 only at the boundary
RTK GNSS (local base)	0.01-0.05 m	Unaffected	Degraded by shadowing	Moderate	4.5	Meets requirement with margin, when satellites are visible
Network RTK / CORS	0.02-0.05 m	Unaffected	Requires connectivity	Moderate	4.0	Good; connectivity dependence is a constraint at Bailadila
PPP / PPP-RTK	0.05-0.1 m	Unaffected	Long convergence	Moderate	3.5	Convergence time is problematic for intermittent operation
RTK + tactical-grade INS	0.02-0.1 m, coasting through outages	Unaffected	Continuous	Moderate-high	4.8	Recommended
Wheel odometry	Drifts	Unaffected	Continuous	Very low	3.0	Essential fusion input, not standalone
UWB / RF trilateration	0.1-0.5 m	Unaffected	Requires dense anchors	Moderate	3.2	Anchor relocation burden on a changing road
Radar / LiDAR odometry and SLAM	0.1-0.5 m	Radar: unaffected	Continuous	Software cost	3.9	Valuable backup layer; radar-based SLAM works in fog
Magnetic / inertial heading	1-3 deg	Unaffected	Unreliable in haematite	Low	1.5	Rejected for heading - see C-06

Recommendation. RTK GNSS with local base stations, dual-antenna for heading, tightly coupled with a tactical-grade IMU and wheel odometry, with radar odometry as a further aiding source. This combination satisfies PR-08 (sub-0.3 m), PR-09 (continuity through outages), PR-10 (sub-degree heading without magnetic reference) and PR-11 (integrity estimate) simultaneously, and is exactly the architecture proven at scale in autonomous haulage.

Dual-antenna heading is specifically required rather than optional: magnetometer-based heading in a high-grade haematite environment with massive steel vehicle bodies is unreliable, and heading error directly corrupts lane guidance at the ranges that matter.

Infrastructure requirement. A local RTK base network with sufficient baseline coverage over the Bailadila ridge and both complexes, with correction delivery over the site RF network. Base station siting must account for ridge geometry; multiple bases may be required.

4. Layer 3 - Communications

Bearer	Latency	Range	Density handling	Infrastructure need	Fog	Score	Verdict
ITS-G5 / DSRC (802.11p)	1-10 ms	300-1000 m LOS	Degrades under high load; CSMA contention	None for V2V	Unaffected	4.2	Strong candidate; direct, infrastructure-free

Bearer	Latency	Range	Density handling	Infrastructure need	Fog	Score	Verdict
C-V2X PC5 direct (LTE-V2X / NR-V2X)	1-20 ms	300-1500 m	Better than DSRC under load; scheduled access	None for direct mode	Unaffected	4.6	Recommended primary
Private LTE / 5G (Uu)	20-100 ms	Cell dependent	Excellent	Full network required	Unaffected	4.0	Recommended for V2I backhaul
Wi-Fi mesh (802.11)	10-50 ms, variable	100-300 m	Moderate	Access point network	Unaffected	3.4	Common in mining; roaming and latency are weaknesses
LoRaWAN / sub-GHz LPWAN	Seconds	Kilometres	Low rate	Gateways	Unaffected	2.5	Suitable for telemetry only, not safety
Proprietary sub-GHz RF (CAS-type)	10-50 ms	~800 m, NLOS tolerant	Good	None	Unaffected	4.4	Proven in mining CAS; non-line-of-sight capability is valuable on switchbacks
VHF/UHF voice	Human latency	Kilometres	Single channel	Repeaters	Unaffected	1.5	Retain as fallback, not as a system layer
Satellite	Seconds	Global	Low	None on site	Unaffected	1.5	Not applicable

Recommendation. A two-tier architecture:

? **Tier 1 - direct V2V** on C-V2X PC5 direct mode or a proven proprietary sub-GHz CAS bearer, satisfying PR-13 (sub-100 ms) and PR-14 (10 Hz), operating **independently of any infrastructure**, satisfying CR-06. C-V2X PC5 is preferred on the merits - better congestion behaviour under the 60-vehicle density of CR-03, a clear evolution path to NR-V2X, and an automotive ecosystem driving cost down - but the proprietary sub-GHz option deserves serious evaluation specifically because of its documented non-line-of-sight performance, which matters on a switchback where two trucks 80 m apart are separated by a rock face. **This choice should be resolved by on-site RF propagation measurement at Bailadila, not by specification comparison.**

? **Tier 2 - V2I backhaul** on private LTE/5G or the existing site network, carrying road model updates, hazard markers, dispatch instructions, telemetry and health reporting, with store-and-forward tolerance of the RF shadows that Bailadila's terrain guarantees.

Spectrum authorisation in India must be confirmed early for whichever bearer is selected; this is a long-lead item and a programme risk if left late.

5. Layer 4 - Compute and Edge Processing

Option	Suitability
Automotive-grade embedded SoC with NPU	Recommended. Deterministic, rugged, low power, suitable temperature range, appropriate cost, and a safety-certification lineage. Adequate for radar fusion, tracking and rendering.
Industrial embedded GPU module	Viable where heavier neural inference is needed for camera and thermal classification. Higher power and thermal load in a hot cab.
Consumer or general-purpose PC hardware	Rejected. Fails the vibration, temperature and ingress requirements ER-01 to ER-04, and introduces rotating or unqualified storage contrary to ER-04.

Option	Suitability
Cloud-based inference	Rejected for safety functions. Violates PR-12 latency and CR-06 infrastructure independence. Appropriate only for offline analytics, model training and fleet reporting.
Roadside edge nodes	Useful for zone visibility sensing and aggregation; must meet the 7-day solar autonomy of ER-08.

Architectural principle. All safety-critical processing - detection, tracking, threat assessment, alerting, lane guidance - occurs **on the vehicle**. The control room receives a picture and issues coordination instructions but is never in the safety loop. This satisfies CR-06 and F6, and it means an RF shadow degrades coordination without degrading protection.

6. Layer 5 - Human-Machine Interface

Option	Assessment
In-cab plan-view display	Recommended primary. Bird's-eye view of host vehicle, road corridor from the digital model, and surrounding contacts. Directly satisfies HR-01 spatial pre-attentive requirement. Proven in mining CAS and maritime ARPA.
Synthetic perspective ("synthetic vision") view	Recommended secondary. Forward perspective view of the road ahead drawn from the digital road model and precise position, with sensed hazards overlaid. Directly transfers the aviation SVS concept. Intuitive for an operator who is trying and failing to see the same scene.
Head-up display on windscreen	Attractive - keeps eyes forward - but complex to retrofit across a heterogeneous fleet, and windscreen geometry varies by machine. Candidate for a later phase.
Directional audio	Recommended. Bearing-encoded alerts satisfy HR-04 and reduce visual load.
Haptic seat or control feedback	Recommended for highest alert level per HR-05; the modality least susceptible to masking.
Text and numeric readouts	Minimise. Violates HR-02 and consumes glance time. Acceptable only for non-safety status.
Augmented reality headset	Rejected. Unsuitable for PPE compatibility, comfort over a full shift, ruggedness and cost.

Recommended interface. Plan view as the primary, with a synthetic perspective view available and automatically promoted in classes V3-V4, directional audio for alerting, and haptic for the highest level. System health, mode and effective sensing range persistently displayed per FR-27 and HR-11.

7. Layer 6 - Visibility Sensing and Nowcasting

Option	Assessment
Forward-scatter visibility meter	Recommended. The standard instrument for calibrated MOR. Modest cost, low power, suitable for solar-powered roadside nodes. Several units required to capture elevation stratification per FR-02.
Transmissometer	Higher accuracy, higher cost, requires a fixed baseline. Justified for one or two reference installations, particularly for acceptance testing per AR-02.

Option	Assessment
Camera-based visibility estimation	Low cost; can be derived from existing CCTV. Less accurate and requires calibration against a reference instrument. Useful for densifying a network whose backbone is proper instruments.
Radar-derived visibility proxy	Research-grade; not recommended as a primary measure.
Vehicle-mounted visibility estimation	Attractive - every truck becomes a mobile sensor, giving dense spatial coverage of a patchy phenomenon at no marginal hardware cost if derived from existing cameras. Recommended as a Phase 2 enhancement.
Numerical weather prediction downscaling	Useful at the multi-hour horizon for shift planning; insufficient skill at the 15-30 minute horizon required by FR-04.
ML nowcasting on local sensor history	Recommended for FR-04. Trained on the site's own accumulating record of visibility, temperature, dew point, humidity, wind and rainfall. Skill improves as the record lengthens, which argues for beginning data collection immediately, ahead of the main deployment.

Visibility sensing is the cheapest layer in the entire architecture and disproportionately valuable. It converts the operating mode decision from subjective to instrumented, enables per-zone rather than mine-wide declarations, provides the independent measurement required for acceptance testing, and builds the dataset on which nowcasting skill depends. **It should be deployed first, ahead of any vehicle equipment**, because it starts accumulating the training and baseline data that everything downstream needs.

8. Recommended Technology Set

Layer	Selected technology	Key requirement satisfied
Cooperative awareness	RF cooperative broadcast and ranging; C-V2X PC5 direct or proven sub-GHz CAS bearer, resolved by on-site RF survey	FR-18 to FR-20, CR-01 to CR-03, PR-13, PR-14
Non-cooperative sensing	4D imaging radar 77/79 GHz, forward long-range plus quarter coverage	FR-11 to FR-15, PR-01 to PR-07
Blind-zone protection	Short-range radar; magnetic-field PDS for personnel, subject to on-site ferrous validation	FR-13, PR-03
Secondary classification	LWIR thermal forward; visible cameras forward and rear	FR-14, GR-02 rear-view compliance
Positioning	RTK GNSS, dual-antenna heading, tightly coupled tactical IMU and odometry, radar odometry aiding	PR-08 to PR-11, FR-05
Road model	Survey-grade LiDAR mapping in clear conditions; field-updatable model with validity timestamp	FR-06, FR-07
Backhaul	Private LTE/5G or existing site network, store-and-forward	CR-05, OR-01 to OR-07
Vehicle compute	Automotive-grade embedded SoC with NPU, all safety processing on-vehicle	PR-12, CR-06, ER-02 to ER-05
Interface	Plan view primary, synthetic perspective secondary, directional audio, haptic for highest alert	HR-01 to HR-11
Visibility	Forward-scatter meter network by elevation zone, one reference transmissometer, ML nowcasting	FR-01 to FR-04, AR-02

9. Technologies Deliberately Rejected

Stating rejections explicitly prevents them re-entering the design later through vendor advocacy.

Technology	Reason for rejection
Visible camera as a primary sensing layer	Physics. Blind at MOR 3-5 m. Retained only as a secondary V0-V2 layer.
NIR/SWIR LiDAR as a real-time fog sensor	Physics. Range collapses in dense fog. Retained for offline mapping only.
Thermal imaging as a <i>sole</i> fog solution	Partial penetration only; poor on thermally neutral targets; fouling aperture. Retained as secondary.
Standard GNSS or SBAS for lane guidance	Accuracy coarser than available lane width.
Magnetometer-based heading	Unreliable in high-grade haematite and adjacent to massive steel bodies.
Cloud inference in the safety loop	Latency and infrastructure dependence.
Consumer-grade compute hardware	Environmental qualification failure.
AR headsets	PPE compatibility, shift-long comfort, ruggedness, cost.
Tag-only proximity systems as the complete solution	Blind to all non-cooperative hazards, which increase during monsoon.
Full autonomous haulage as the near-term answer	Economics, terrain, unproven dense-fog envelope, undeveloped Indian regulatory pathway.
Dense fixed roadside sensing infrastructure	Relocation burden on a road network that changes continuously with mine development.

10. Cost and Maturity Positioning

Indicative relative positioning to inform phasing. These are order-of-magnitude indications for comparison only and must be replaced by quotations during procurement.

Component	Relative unit cost	Technology readiness	Phase
Visibility sensor node (forward-scatter, solar)	Low	Very high	1
Cooperative RF unit per vehicle	Low	Very high	1
In-cab display and compute	Low-moderate	Very high	1
RTK base infrastructure	Moderate, fixed	Very high	1
GNSS/INS per vehicle	Moderate	Very high	2
4D imaging radar per vehicle	Moderate	High	2
Short-range radar set per vehicle	Low	Very high	2
LWIR thermal per vehicle	Moderate-high	High	3
Digital road model creation and maintenance	Moderate, recurring	High	2
Control room integration	Moderate, fixed	High	1-2
Nowcasting model development	Low, software	Moderate	3
Intervention capability and DGMS approval	High, long lead	Moderate	4

Phase 1 - visibility sensing, cooperative awareness, positioning infrastructure and the control room picture - delivers a substantial fraction of the value at the lowest cost and technical risk, and establishes the measurement baseline against which later phases are justified. Phase 2 adds the non-cooperative radar layer and full lane guidance, which is where the V3 operating class becomes defensible. This phasing is developed in Volume II Document 3.